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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)

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to check your
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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1
Introduction
and
Principles of
Algorithms ()

Week 2
Shortest
Path
Algorithms
and
problems ()

☐ Algorithms for
finding
Shortest Path
(unit?)

Week 2 Assignment 2

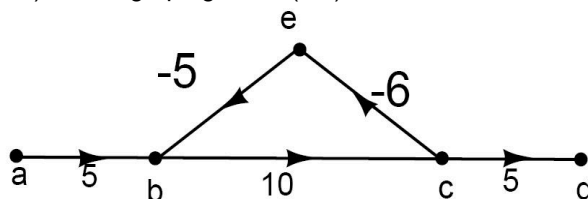
The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

As per our records you have not submitted this assignment.

1) In the graph given, $\alpha(a,b)=$

2 points



☐ ∞

☐ $-\infty$

☐ 0

☐ 5

No, the answer is incorrect.

Score: 0

Accepted Answers:

$-\infty$

2) State True or False

1 point

For any directed graph G , $\alpha(u,v) \leq \delta(u,v)$ for any pair of vertices u and v of G .

☐ True

☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

True

unit=19&lesso
n=38)

☐ Algorithms for
finding
Shortest Path
Part 2 (unit?
unit=19&lesso
n=39)

☐ Single source
shortest path
problem (unit?
unit=19&lesso
n=40)

☐ Properties of
shortest path
distances
(unit?
unit=19&lesso
n=41)

☐ Properties of
shortest path
distances Part
2 (unit?
unit=19&lesso
n=42)

☐ Week 2
Lecture
Material (unit?
unit=19&lesso
n=51)

☐ Quiz: Week 2
Assignment 2
(assessment?
name=98)

☐ Feedback
Form (unit?
unit=19&lesso
n=27)

Week 3
Bellman
Equations ()

Week 4
Bellman
Ford and
Dijkstra
Algorithm ()

Week 5 All
Pair Shortest

3) State True or False

Let G be a graph with no negative cycles then $\alpha(u, v) < \delta(u, v)$

- ☐ True
☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

False

4) Let $G=(V, E, W, s)$ be an instance of a SSSP Problem. Let $\delta(v)$ denote the weight of the shortest path from s to v . Write down Bellman Equation satisfied by values $\delta(u), u \in V$. **3 points**

☐
 $\delta(s) = 0 \text{ and } \delta(u) = \underset{v \neq u}{Min} \{ \delta(v) + w(v, u) \} \text{ for } u \neq s, u \in V$

☐
 $\delta(s) = 1 \text{ and } \delta(u) = \underset{v \neq u}{Min} \{ \delta(v) + w(v, u) \} \text{ for } u \neq s, u \in V$

☐
 $\delta(s) = 0 \text{ and } \delta(u) = \underset{v = u}{Min} \{ \delta(v) + w(v, u) \} \text{ for } u \neq s, u \in V$

☐
 $\delta(s) = 0 \text{ and } \delta(u) = \underset{v \neq u}{Min} \{ \delta(v) + w(u, v) \} \text{ for } u \neq s, u \in V$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\delta(s) = 0 \text{ and } \delta(u) = \underset{v \neq u}{Min} \{ \delta(v) + w(v, u) \} \text{ for } u \neq s, u \in V$

1 point



Path ()

**Week 6
Prims and
Kruskal's
Algorithm ()**

**Week 7 DFS
and Cut
Vertex ()**

**Video
download ()**

**Live session
()**

